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SMART GARDENING · AGRONOMY REFERENCE FOR THE ML TEAM

Irrigation Simulator Handbook

The physics, formulas and parameter values you need to simulate hourly soil moisture in irrigated ground beds. It's built on the FAO-56 soil water balance, the standard method in irrigation science, adapted to an hourly step.

Every formula comes with what it means physically, its units, and typical values. Tabulated values are starting points. Real soils and plants vary, so plan to calibrate against sensor data (section 12).

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00 How the simulator works

Think of the root zone as a bucket of soil. Water comes in from rain and irrigation. It leaves through four paths: evaporation from the soil surface, transpiration through the plant, drainage below the roots, and runoff across the surface. Each hour the simulator does this:

1. Compute how much water the atmosphere *demands* from the weather (reference evapotranspiration, ET_0).

2. Scale that demand to the actual plant and its growth stage (crop coefficients).
3. Reduce it when the soil is too dry for the plant to keep up (water stress).
4. Add rain and irrigation, minus what runs off or is intercepted by leaves.
5. Drain any water above field capacity slowly downward.
6. Report what a real moisture sensor would read, including noise and faults.

Notation used throughout

θ	Volumetric water content, m^3 water per m^3 soil (a fraction; 0.25 = 25 % vol).
Z_r	Effective root depth, m.
W	Water stored in the root zone, mm. $W = 1000\,\theta\,Z_r$.
D_r	Root-zone depletion, mm below field capacity. $D_r = W_{FC} - W$.
ET_0, ET_c	Reference and crop evapotranspiration, mm per hour.
P, I, RO, DP	Precipitation, irrigation, runoff, deep percolation (drainage), mm per hour.
i	Hour index. Hourly quantities are totals over the hour.

Unit you'll use everywhere.

1 mm of water over 1 m^2 is exactly 1 litre. Hydrologists work in mm because it doesn't depend on area. Convert to litres only when you size the valve run.

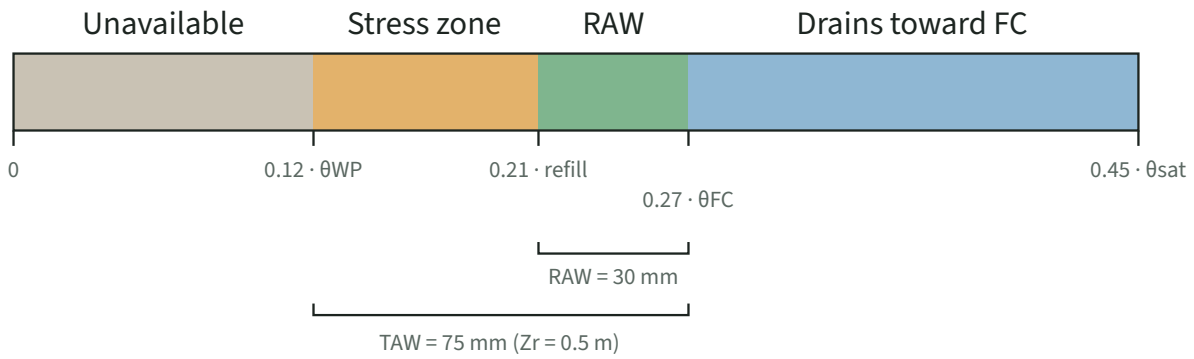
01 Soil water storage

Soil holds water in the pores between particles. Four moisture levels define how that water behaves:

θ_{sat}	**Saturation.** Every pore is full. Roughly equal to the soil's porosity.
θ_{FC}	**Field capacity.** What remains once gravity has drained the big pores, typically 1–3 days after a soaking. Water above FC doesn't stay; it drains.

θ_{refill} **Refill point** (management threshold). Below this the plant starts to struggle to extract water and closes its stomata. This is where you want to irrigate.

θ_{WP} **Permanent wilting point.** Water is held so tightly (about −1.5 MPa suction) that roots can't extract it. Plants wilt and don't recover overnight.



Water zones for a loam with 0.5 m roots and p = 0.4. Keeping θ between the refill point and field capacity is the agronomic target band.

Total and readily available water

$$TAW = 1000\,(\theta_{FC} - \theta_{WP})\,Z_r \qquad [\mathrm{mm}] \tag{E1.1}$$

$$RAW = p \cdot TAW, \qquad \theta_{refill} = \theta_{FC} - \frac{RAW}{1000\,Z_r} \tag{E1.2}$$

TAW (total available water) is everything a plant could possibly extract. **RAW** (readily available water) is the part it extracts with no stress. **p** is the depletion fraction, a crop property (section 3): shallow-rooted, sensitive crops like lettuce have p ≈ 0.3, and tougher crops have 0.5 or more. Irrigation engineers call p the *management allowed depletion*.

Typical soil properties by texture

Typical soil properties by texture

Texture	θ_{FC}	θ_{WP}	TAW per m root	θ_{sat}	K_{sat} mm/h	REW mm	Intake r
Sand	0.07–0.17	0.02–0.07	~75 mm	0.44	210	2–7	> 30
Loamy sand	0.11–0.19	0.03–0.10	~85 mm	0.44	61	4–8	25–30*
Sandy loam	0.18–0.28	0.06–0.16	~120 mm	0.45	26	6–10	20–30
Loam	0.20–0.30	0.07–0.17	~130 mm	0.46	13	8–10	10–20
Silt loam	0.22–0.36	0.09–0.21	~140 mm	0.50	6.8	8–11	10–15*
Silty clay loam	0.30–0.37	0.17–0.24	~130 mm	0.47	1.5	8–11	5–10*
Silty clay	0.30–0.42	0.17–0.29	~130 mm	0.48	0.9	8–12	1–5*
Clay	0.32–0.40	0.20–0.24	~140 mm	0.48	0.6	8–12	1–5

θ_{FC}, θ_{WP}, REW: FAO-56 Table 19. θ_{sat} (total porosity) and K_{sat}: Rawls, Brakensiek & Saxton (1982) class means. Intake rate: FAO basic infiltration classes. * marks values interpolated between FAO classes. Garden soils amended with compost usually hold more water than these mineral-soil values suggest.

For the model.

θ_{FC}, θ_{WP} and Z_r are the three parameters that matter most. The ranges are wide, so treat them as priors and fit them per bed from sensor data (section 12). The ±0.05 spread on θ_{FC} alone moves TAW by about 50 %.

Root depth

Use the effective rooting depth, capped by whatever limits it: $Z_r = \min(Z_{r,crop},\ \text{soil depth})$. A bed over compacted subsoil or rubble effectively ends at that layer. Roots grow during the season, and a linear ramp is good enough:

$$Z_r(t) = Z_{r,min} + (Z_{r,max} - Z_{r,min})\,\frac{t - t_0}{t_{max} - t_0}, \qquad Z_{r,min} \approx 0.10\text{–}0.15\ \mathrm{m} \tag{E1.3}$$

Here t_0 is planting and t_{max} is the start of the mid-season stage. When Z_r grows, newly explored soil brings its own water with it. Simulate that by adding the water held in the new depth increment at that layer's θ.

02 Reference evapotranspiration (ET_0)

ET_0 is the water a well-watered, 12 cm tall grass lawn would lose in a given hour. It depends only on weather, which makes it a single number for the "drying power" of the air. Every plant's water use is then expressed relative to it.

Shortcut.

Open-Meteo returns `et0_fao_evapotranspiration` hourly, computed with this same equation. You can use it directly and implement the formulas below only to validate it or to compute ET_0 from your own station. When you simulate the past, compute it from *observed* weather. When the controller plans, use the *forecast* ET_0.

Hourly FAO-56 Penman-Monteith

$$ET_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\dfrac{37}{T_{hr} + 273}\,u_2\,(e^{\circ}(T_{hr}) - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \qquad [\mathrm{mm\ h^{-1}}] \tag{E2.1}$$

The numerator has two drivers. The first term is **energy**: radiation evaporates

The hammer has two drivers. The first is energy: radiation evaporates water. The second is **aerodynamics**: dry, windy air carries vapour away. A hot, sunny summer midday gives about 0.5–0.9 mm/h. At night ET_0 is close to zero.

T_{hr} Mean air temperature over the hour, °C (2 m height).

u_2 Wind speed at 2 m, m/s.

R_n Net radiation at the surface, MJ m^{-2} h^{-1}.

G Soil heat flux, MJ m^{-2} h^{-1}. $G = 0.1\,R_n$ when $R_n > 0$ (day), $G = 0.5\,R_n$ at night.

Supporting equations

$$e^\circ(T) = 0.6108\,\exp\!\left(\frac{17.27\,T}{T + 237.3}\right) \quad [\text{kPa}] \qquad e_a = e^\circ(T_{hr})\,\frac{RH_{hr}}{100}$$

Saturation vapour pressure, and actual vapour pressure from relative humidity. The gap $e^\circ - e_a$ (vapour pressure deficit, VPD) measures how "thirsty" the air is.

$$\Delta = \frac{4098\,e^\circ(T_{hr})}{(T_{hr} + 237.3)^2} \qquad \gamma = 0.665 \times 10^{-3}\,P \qquad P = 101.3\left(\frac{293 - 0.0065\,z}{293}\right)^{5.26}$$

Slope of the vapour pressure curve (kPa/°C), psychrometric constant (kPa/°C), and atmospheric pressure (kPa) from elevation z in metres. You can also use measured surface pressure.

$$u_2 = u_z\,\frac{4.87}{\ln(67.8\,z - 5.42)} \qquad (\text{for } z = 10\text{ m: } u_2 = 0.748\,u_{10})$$

Wind is usually reported at 10 m. This log-profile converts it to 2 m.

Net radiation

$$R_n = R_{ns} - R_{nl}, \qquad R_{ns} = (1 - 0.23)\, R_s$$

$$R_{nl} = \sigma\,(T_{hr} + 273.16)^4\,(0.34 - 0.14\sqrt{e_a})\left(1.35\,\frac{R_s}{R_{so}} - 0.35\right), \qquad \sigma = 2.043 \times 10^{-1}$$

R_s is measured incoming shortwave radiation, and 0.23 is the albedo of the grass reference. R_{nl} is the longwave heat the ground radiates back to the sky, which is lower under clouds. The ratio R_s/R_{so} is a cloudiness index: keep it between 0.3 and 1.0. At night it's undefined, so reuse the value from 2–3 hours before sunset.

$$R_{so} = (0.75 + 2 \times 10^{-5}\, z)\, R_a$$

Extraterrestrial radiation, hourly

$$R_a = \frac{12 \cdot 60}{\pi}\, G_{sc}\, d_r\, [(\omega_2 - \omega_1)\sin\varphi\sin\delta + \cos\varphi\cos\delta\,(\sin\omega_2 - \sin\omega_1)], \qquad G_{sc} = 0$$

$$d_r = 1 + 0.033\cos\left(\frac{2\pi J}{365}\right) \qquad \delta = 0.409\sin\left(\frac{2\pi J}{365} - 1.39\right)$$

$$\omega = \frac{\pi}{12}[(t + 0.06667\,(L_z - L_m) + S_c) - 12], \qquad \omega_{1,2} = \omega \mp \frac{\pi}{24}$$

$$S_c = 0.1645\sin 2b - 0.1255\cos b - 0.025\sin b, \qquad b = \frac{2\pi(J - 81)}{364}$$

J is day of year, φ is latitude in radians, and t is the clock time at the *midpoint* of the hour (14.5 for 14:00–15:00). Clip ω_1 and ω_2 to the sunset hour angle

$\omega_s = \arccos(-\tan\varphi \tan\delta)$; when the sun is below the horizon, $R_a = 0$.

Sign trap.

FAO-56 measures L_z (time-zone centre) and L_m (site longitude) in degrees *west* of Greenwich, from 0 to 360°. Madrid (3.7° W) has $L_m = 3.7$, and Berlin (13.4° E) has $L_m = 346.6$. The simplest safe approach is to run everything in UTC with $L_z = 0$, and keep daylight saving time out of it.

03 Crop coefficients

A tomato plant doesn't use the same water as the reference grass. A crop coefficient scales ET_0 to the real plant, and it changes as the plant grows. There are two levels of detail:

$$\underbrace{ET_c = K_c\, ET_0}_{\text{single coefficient}} \qquad \underbrace{ET_c = (K_s K_{cb} + K_e)\, ET_0}_{\text{dual coefficient}} \tag{E3.1}$$

The single K_c lumps plant transpiration and soil evaporation together. The dual form separates $\mathbf{K_{cb}}$ (basal coefficient: transpiration from the plant) from $\mathbf{K_e}$ (evaporation from wet soil, section 4). Use the dual form for an hourly simulator. Right after rain or irrigation, the wet soil surface evaporates fast for a day or two, and only the dual form captures that spike. K_s (section 5) reduces transpiration when the soil is dry.

Growth stages

FAO-56 divides the season into four stages: **initial** (planting to ~10 % ground cover), **development** (to full cover), **mid-season** (full cover to start of maturity)

and **late season** (to harvest). K_{cb} is constant in the initial and mid stages and ramps linearly during development and late season:

$$K_{cb}(d) = K_{cb,prev} + \frac{d - \sum L_{prev}}{L_{stage}} \left(K_{cb,next} - K_{cb,prev}\right) \quad \text{E3.2}$$

Crop	K_{cb} ini / mid / end	K_c ini / mid / end	Max root depth m	p	Max I
Tomato	0.15 / 1.10 / 0.70–0.90	0.60 / 1.15 / 0.70–0.90	0.7–1.5	0.40	0.6
Sweet pepper	0.15 / 1.00 / 0.80	0.60 / 1.05 / 0.90	0.5–1.0	0.30	0.7
Lettuce	0.15 / 0.90 / 0.90	0.70 / 1.00 / 0.95	0.3–0.5	0.30	0.3
Spinach	0.15 / 0.95 / 0.85	0.70 / 1.00 / 0.95	0.3–0.5	0.20	0.3
Green bean	0.15 / 1.00 / 0.80	0.50 / 1.05 / 0.90	0.5–0.7	0.45	0.4
Carrot	0.15 / 0.95 / 0.85	0.70 / 1.05 / 0.95	0.5–1.0	0.35	0.3
Cucumber	0.15 / 0.95 / 0.70	0.60 / 1.00 / 0.75	0.7–1.2	0.50	0.3
Potato	0.15 / 1.10 / 0.65	0.50 / 1.15 / 0.75	0.4–0.6	0.35	0.6
Strawberry	0.30 / 0.80 / 0.70	0.40 / 0.85 / 0.75	0.2–0.3	0.20	0.2
Lawn, cool-season	0.85 / 0.90 / 0.90	0.90 / 0.95 / 0.95	0.5–1.0	0.40	0.10

From FAO-56 Tables 11, 12, 17 and 22 (sub-humid climate, $RH_{min} \approx 45$ %, $u_2 \approx 2$ m/s). Stage lengths are one example region each and vary a lot with climate and sowing date. Garden crops grown on supports, or harvested continuously, stay in mid-season longer.

Climate adjustment for mid and end K_{cb}

Dry, windy climates raise K_{cb}; humid, calm ones lower it. Apply this to values

above 0.45:

$$K_{cb} = K_{cb(tab)} + [0.04\,(u_2 - 2) - 0.004\,(RH_{min} - 45)]\left(\frac{h}{3}\right)^{0.3} \quad \text{E3.3}$$

u_2 is the mean daily wind and RH_{min} is the mean daily minimum relative humidity over the stage. h is plant height in metres.

Hourly use.

FAO-56 coefficients were derived on a daily basis. Applying them to hourly ET_o and summing reproduces daily ET_c well enough for this purpose. Treat K_{cb} as constant within a day.

04 Soil evaporation

A freshly wetted soil surface evaporates almost like open water. As the top 10–15 cm dries, evaporation slows sharply. FAO-56 models this with a thin surface layer of depth Z_e and its own depletion D_e.

$$K_e = \min(K_r\,(K_{c,max} - K_{cb}),\;\; f_{ew}\,K_{c,max}) \quad \text{E4.1}$$

$$K_{c,max} = \max\left(1.2 + [0.04(u_2 - 2) - 0.004(RH_{min} - 45)]\left(\frac{h}{3}\right)^{0.3},\;\; K_{cb} + 0.05\right)$$

$K_{c,max}$ is the ceiling on total ET from a fully wet surface.

Stage 1 and stage 2 drying

$$K_r = \begin{cases} 1 & D_e \le REW \\ \dfrac{TEW - D_e}{TEW - REW} & D_e > REW \end{cases} \qquad TEW = 1000\,(\theta_{FC} - 0.5\,\theta_{WP})\,Z_e \tag{E4.2}$$

REW (readily evaporable water) evaporates at the full rate (stage 1). After that, the rate falls linearly until **TEW** (total evaporable water) has gone (stage 2). Use Z_e = 0.10–0.15 m, and REW from the soil table.

Where the soil is exposed and wet

$$f_{ew} = \min(1 - f_c,\ f_w) \qquad f_c = \left(\frac{K_{cb} - K_{c,min}}{K_{c,max} - K_{c,min}}\right)^{1+0.5h},\ \ K_{c,min} \approx 0.15$$

f_c is the fraction of ground shaded by the canopy, estimated from K_{cb} when not measured. f_w is the fraction of the surface wetted: 1.0 for rain and sprinklers, 0.3–0.4 for drip, and 0.4–0.6 for a micro-sprayer. Drip under a dense canopy means little soil evaporation, which is part of why drip is efficient.

Surface layer balance (hourly)

$$D_{e,i} = \mathrm{clip}\left(D_{e,i-1} - (P_i - RO_i) - \frac{I_i}{f_w} + \frac{E_i}{f_{ew}},\ 0,\ TEW\right), \qquad E_i = K_e\,I$$ E4.3

This is FAO-56 eq. 77 with two small terms dropped: transpiration drawn from the surface layer, and drainage out of it, which the lower clip handles. Irrigation is divided by f_w because drip water is concentrated on the wetted spots.

05 Water stress

When depletion exceeds RAW, the plant can no longer pull water fast enough and partly closes its stomata. Transpiration (and growth) drop roughly linearly to zero at the wilting point:

$$K_s = \begin{cases} 1 & D_r \le RAW \\ \dfrac{TAW - D_r}{(1-p)\,TAW} & RAW < D_r < TAW \\ 0 & D_r \ge TAW \end{cases} \qquad \text{E5.1}$$

p depends on how hard the atmosphere is pulling. On hot days the plant can't keep up even in moister soil. FAO-56 adjusts the table value with daily crop ET:

$$p = \mathrm{clip}(p_{tab} + 0.04\,(5 - ET_{c,day}),\ 0.1,\ 0.8) \qquad [ET_{c,day} \text{ in mm/day}] \qquad \text{E5.2}$$

In an hourly model, use the rolling 24-hour sum of ET_c for $ET_{c,day}$.

For the model.

K_s is your continuous, physically grounded stress signal: 1 means no stress and 0 means wilting. It's a latent variable. The simulator knows it, the field sensor doesn't, so use it to *score* the controller and never as a model input (section 10).

Too wet is also stress

Roots need oxygen. When almost all pores are full of water, roots suffocate within

hours to days, and root rot follows. A widely used agronomic threshold is **air-filled porosity below 10 %**:

$$\text{waterlogged if } \theta > \theta_{sat} - 0.10 \qquad \text{E5.3}$$

FAO-56 has no formula for this. Count hours above the threshold as a separate penalty. Short events right after heavy rain are normal; sustained ones mean overwatering or poor drainage.

06 Rain, interception and runoff

Canopy interception

Leaves catch a thin film of water that evaporates without reaching the soil. Model it as a small bucket with capacity proportional to leaf area:

$$S_{can} \approx 0.1\text{–}0.2\ \text{mm} \times LAI, \qquad P_{net} = P - \min(P,\ S_{can} - S_{stored})$$

The intercepted water then evaporates at the reference rate. For vegetable beds (leaf area index LAI ≈ 1–4) this is a fraction of a millimetre per event. Include it for completeness, but it's minor.

Small rain doesn't count

FAO-56 notes that daily rain below about $0.2 \times ET_0$ normally evaporates entirely from the surface. The dual-coefficient model handles this automatically: small rain only refills D_e and leaves through K_e. This matters for the controller. A forecast of 1 mm of drizzle shouldn't cancel irrigation.

Runoff, hourly

Water runs off when it arrives faster than the soil can absorb it (infiltration excess), or when the soil is already full (saturation excess):

$$RO_i = \max(0,\ P_{net,i} + I_i - f_{cap}) + \max(0,\ W_i^* - W_{sat}), \qquad f_{cap} \approx c \cdot K_{sa} \quad \text{E6.1}$$

W* is storage after adding the infiltrated water. Dry soil absorbs faster than K_{sat} because suction pulls water in. Use c ≈ 3 when dry and c → 1 near saturation, for example $c = 1 + 2\,D_r/TAW$. Ground beds with loose, mulched soil rarely produce runoff except in clay during intense storms. Flat beds can pond instead of running off; you can store the excess and let it infiltrate the next hour.

Alternative.

The USDA **curve number** method, $Q = (P - 0.2S)^2/(P + 0.8S)$ with $S = 25400/CN - 254$ mm, is standard for daily catchment runoff but was designed for storm totals. For hourly bed-scale simulation, the infiltration-capacity form above is a better fit.

Capillary rise

A shallow water table (under about 1–2 m) can feed roots from below. Urban garden beds almost never have one, so set CR = 0 unless you know otherwise.

07 Root-zone balance and drainage

The core equation is plain conservation of mass. Writing it in storage W (mm) rather than depletion makes it easy to represent water above field capacity:

$$W_i = W_{i-1} + (P_{net,i} - RO_i) + I_{net,i} + CR_i - E_i - T_i - DP_i \tag{E7.1}$$

$$T_i = K_s\, K_{cb}\, ET_{0,i} \qquad E_i = K_e\, ET_{0,i} \qquad \theta_i = \frac{W_i}{1000\, Z_r}$$

Drainage: don't drain instantly

The textbook FAO-56 bucket sends everything above field capacity down within the same day. Hourly, that's wrong, and your sensors will show it: after a soaking, θ stays above FC for many hours to a few days while it drains. Use the AquaCrop drainage characteristic, which drains faster when wetter:

$$\frac{\Delta\theta}{\Delta t} = \tau\,(\theta_{sat} - \theta_{FC})\,\frac{e^{\,\theta-\theta_{FC}} - 1}{e^{\,\theta_{sat}-\theta_{FC}} - 1}, \qquad \tau = 0.0866\, K_{sat}^{0.35}\ \ [\mathrm{day}^{-1}],\ K_{sat}\ \text{in} \tag{E7.2}$$

τ is the fraction of the excess that drains in one day. A linear reservoir is a close, simpler alternative:

$$DP_i = k_h\ \max(0,\ W_{i-1} - W_{FC}), \qquad k_h = 1 - (1-\tau)^{1/24} \tag{E7.3}$$

Typical τ values from the formula are about 0.8 for sandy loam, 0.65 for loam and 0.2 for clay. Sand gives above 1 and is capped at 1 (excess drains within a day). Fit k_h from the recession curves in your own sensor data after rain; it's one of the easiest parameters to identify.

Bounds

- $W \leq W_{sat}$. Anything above becomes runoff or ponding (E6.1).
- $W \geq 0$. In practice $K_S \to 0$ at W_{WP} stops transpiration, and only surface evaporation can dry the soil slightly below wilting point.

08 Irrigation dose and duration

This is the "physics instead of learning" step from the architecture discussion. The forest predicts where θ is heading. Once the behavior tree decides to irrigate, the dose follows from the water balance.

$$I_n = \lambda D_r = \lambda \cdot 1000 \left(\theta_{FC} - \theta\right) Z_r \qquad [\mathrm{mm}]$$ E8.1 net depth

λ is the fraction of the deficit to refill. λ = 1 refills to field capacity. λ ≈ 0.7–0.9 leaves headroom for rain that might arrive. Refilling *above* FC wastes water, because the excess drains away.

$$I_g = \frac{I_n}{E_a} \qquad V = I_g \cdot A \qquad t_{run} = \frac{V}{Q}$$ E8.2 gross, litres, hours

E_a Application efficiency: the fraction of delivered water that ends up in the root zone. Drip 0.85–0.95, micro-sprinkler 0.80–0.90, sprinkler 0.65–0.80, hose or flood 0.50–0.70.

A Bed area in m^2 (mm × m^2 = L).

Q System flow, L/h. For drip, Q = number of emitters × emitter rate (usually 1, 2 or 4 L/h). Measure it once with a bucket and timer, or better, with a flow meter.

Don't apply faster than the soil absorbs

$$Q$$

$$AR = \frac{Q}{A} \ [\text{mm/h}] \qquad \text{require } AR \le \text{intake rate} \tag{E8.3}$$

If the application rate exceeds the soil's intake rate (soil table), water ponds and runs off. The fix is **cycle and soak**: split the run into pulses, each short enough to soak in, with 30–60 minutes between them. On clay you'll almost always need it with sprinklers. Drip rarely exceeds intake, but on clay the wetted spot can still pond.

In the simulator

The water that reaches the root zone is $I_{net} = E_a \cdot V/A$ mm. The loss $(1 - E_a)$ is spray evaporation, drift and deep percolation directly under the emitters. Put drip water on the f_w fraction of the surface (E4.3). Irrigating early morning reduces evaporation and wind losses, which is why E_a for sprinklers is at the high end before 9:00.

Calculator

Soil texture: Loam

θ field capacity: 0.27

θ wilting point: 0.12

Root depth Zr (m): 0.5

Depletion fraction p: 0.4

Current θ (sensor): 0.20

Refill fraction λ: 1

Bed area (m²): 2

Efficiency Ea: 0.9

System flow Q (L/h): 16

TAW RAW Refill point θ Depletion Dr

75 mm	30 mm	0.210	35.0 mm
Stress Ks	Net depth In	Volume V	Run time
0.89	35.0 mm	77.8 L	4 h 52 min
Application rate			
8.0 mm/h			

Soil is below the refill point (Ks = 0.89). Irrigation is warranted unless enough rain is coming.

Example values: a 2 m² loam bed with 8 drippers at 2 L/h. Choosing a texture loads the FAO-56 midpoint θ values. The intake check uses the lower bound of the class range.

09 Sensor model and faults

Your ML model never sees the true root-zone θ. It sees a sensor reading at one point and one depth, through a sensor with its own physics. The simulator should produce the same kind of signal.

Depth matters

A sensor at 10 cm responds within hours to rain and evaporation and swings widely. A sensor at 30 cm is smoother and lags, but represents what most roots experience. A practical mapping from the model's two layers:

$$\theta_{top} \approx \theta_{FC} - \frac{D_e}{1000\, Z_e} \qquad \theta_{deep} \approx \frac{W - 1000\,\theta_{top}\, Z_e}{1000\,(Z_r - Z_e)}$$

Map a 5–10 cm sensor to θ_{top} and a 20–40 cm sensor to θ_{deep}. If you need a finer vertical profile, use a multi-layer model like AquaCrop's, but two layers are usually enough for control.

Observation model

$$\theta_{obs,i} = g(\theta_{layer,i}) + b + \beta t_i + \kappa\,(T_{soil,i} - 20) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \tag{E9.1}$$

- `g(·)` Calibration curve. Capacitive sensors report a raw count that relates nonlinearly to θ, and the curve differs by soil. A generic factory curve is typically off by several % vol, and soil-specific calibration brings it to about ±2 % vol.
- `b` Installation offset: air gaps, stones, or compaction around the probe. Constant per install, ±0.01–0.04.
- `β` Slow drift from probe aging or salt build-up.
- `κ` Temperature sensitivity. Cheap capacitive probes show a visible daily cycle that is not real water movement.
- `σ` Random noise, about 0.003–0.01 m^3/m^3 per reading.

Fault types to inject

Fault	What it looks like	Real-world cause
Dropout	Missing values, single or for hours	Radio loss, battery, gateway dow
Stuck	Exactly constant value for many hours	Frozen firmware, failed ADC
Spike	Single impossible jump, then back	Electrical interference, a waterin can poured right on the probe
Step offset	Sudden permanent shift	Probe knocked, dug around or re inserted
Air gap after drying	Reads too dry and stops responding to small rains	Clay shrinks away from the prob dry spells
Drift	Slow trend unrelated to weather	Corrosion, salt accumulation fro fertilizer

Out of range	θ below 0 or above θ_{sat}	Wiring fault, wrong calibration

Physics checks the behavior tree can use: θ can't rise without rain or irrigation (beyond noise), and it can't fall faster than about ET_c + drainage allows. A drop of 0.05 in one dry hour is a fault, not drying.

10 Plant outcome metrics

These turn the simulator's hidden variables into scores for your evaluation. They are agronomically meaningful, and the simulator computes them for free.

Relative yield from water stress

The FAO yield response relationship (FAO-33, Doorenbos & Kassam) links seasonal water shortfall to yield loss:

$$1 - \frac{Y_a}{Y_m} = K_y \left(1 - \frac{\sum ET_{c,adj}}{\sum ET_c}\right) \quad \text{E10.1}$$

$ET_{c,adj}$ is actual (stressed) ET and ET_c is unstressed. $K_y > 1$ means yield drops proportionally faster than water use. Seasonal values: tomato 1.05, pepper 1.1, bean 1.15, potato 1.1, onion 1.1, cabbage 0.95. Leafy greens are very sensitive to stress for quality (bolting, bitterness), even without a published K_y.

Suggested evaluation set

Metric	Definition	Direction
Time in band	% of hours with $\theta_{refill} \le \theta_{root\ zone} \le \theta_{FC}$	higher

Stress-hours	$\sum_i (1 - K_{s,i})$ over daylight hours	lower
Waterlogging hours	Hours with $\theta > \theta_{sat} - 0.10$ (E5.3)	lower
Relative yield	Y_a/Y_m from E10.1	higher
Water used	Total litres delivered	lower
Drainage losses	$\sum DP$ within 24 h after irrigation events, L	lower
Wasted before rain	Irrigation volume followed by $\geq I_n$ of rain within 24 h	lower
Safety violations	Hours below $\theta_{WP} + 0.02$, or irrigation during a known sensor fault	zero

For the model.

Keep a strict wall between what the simulator knows (true θ, K_s, D_e, drainage, actual future weather) and what production will know (sensor readings, forecasts as issued, valve log). Train and run the forest and behavior tree only on the production-visible signals. Use the hidden ones only for scoring. Leaking a hidden variable gives excellent simulated results that won't transfer.

11 Hourly loop (pseudocode)

The order matters. Inputs arrive first, then losses are computed from the state at the start of the hour, then drainage, then observation.

```
def step(s, wx, irr_litres, prm):
    """s: state, wx: OBSERVED weather this hour, irr_litres: controller output."
    et0 = wx.et0 if wx.et0 is not None else et0_hourly(wx, prm.site)    # §2
    kcb, h = kcb_at(s.day, prm.crop)                                    # §3 E3.2
    kcmax  = kc_max(wx.u2_day, wx.rhmin_day, h, kcb)                    # §4

    # --- inputs ----------------------------------------------------
```

```
    # --- inputs
    p_net   = s.canopy.intercept(wx.precip, s.lai)                    # §6
    i_net   = prm.Ea * irr_litres / prm.area                          # §8, mm
    f_cap   = (1 + 2 * s.Dr_frac()) * prm.Ksat
    ro_inf  = max(0.0, p_net + i_net - f_cap)                         # infiltr
    inflow  = p_net + i_net - ro_inf
    rain_in = inflow * p_net / max(p_net + i_net, 1e-9)
    irr_in  = inflow - rain_in

    # --- surface evaporation ------------------------------------------
    s.De = max(0.0, s.De - rain_in - irr_in / prm.fw)
    kr   = 1.0 if s.De ⩽ prm.REW else max(0.0, (prm.TEW - s.De) / (prm.TEW - prm
    few  = max(0.01, min(1 - canopy_cover(kcb, kcmax, h), prm.fw))
    ke   = min(kr * (kcmax - kcb), few * kcmax)
    E    = ke * et0
    s.De = min(prm.TEW, s.De + E / few)

    # --- transpiration with stress (from state at start of hour) ---------
    p_adj = clip(prm.crop.p + 0.04 * (5 - s.etc_24h()), 0.1, 0.8)
    ks    = water_stress(s.Dr(), s.TAW(), p_adj)                      # §5 E5.1
    T     = ks * kcb * et0

    # --- root zone ----------------------------------------------------
    s.W  += inflow - E - T
    ro_sat = max(0.0, s.W - s.W_sat())                                 # satura
    s.W  -= ro_sat
    dp    = prm.k_h * max(0.0, s.W - s.W_fc())                        # §7 E7.3
    s.W   = max(0.0, s.W - dp)

    s.log(et0=et0, E=E, T=T, ks=ks, dp=dp, ro=ro_inf + ro_sat,
          etc_unstressed=(kcb + ke) * et0)
    s.day_advance_if_needed()
    return observe(s, prm.sensors, s.faults)                          # §9 E9.1
```

Run it for a full season per simulated bed. The controller only ever receives `observe(...)` outputs plus the *forecast* weather for upcoming hours, never `wx` for future hours.

12 Calibration and domain randomization

Calibrate against real stations

Before trusting the simulator, check that it reproduces real soil behaviour. ISMN and USCRN stations have hourly soil moisture at several depths (USCRN: 5, 10, 20, 50, 100 cm) with co-located weather, and no irrigation. That makes them a clean test of the drying and drainage physics:

1. Pick stations whose climate and soil texture resemble your site.
2. Drive the simulator with the station's own weather. Set the crop to the station's ground cover (often grass, $K_{cb} \approx 0.85–0.95$).
3. Fit θ_{FC}, θ_{WP}, Z_r, k_h, REW, and the sensor offset b by minimizing RMSE between simulated and observed θ at matching depths. A small Bayesian fit or CMA-ES works well, since there are only about 6 parameters.
4. Check behaviour, not just RMSE. Does the post-rain recession look right? Does drying speed up on sunny, windy days? Does a 2 mm shower barely register at 20 cm?

Once your own beds have sensors, repeat the fit on them with the irrigation log included. That refines E_a and the dose response as well.

Randomization ranges for training diversity

Parameter	Suggested distribution
Soil texture	Categorical over the soil table, weighted toward loams for garden soil
θ_{FC}, θ_{WP}	Texture midpoint ± uniform 0.03, keeping $\theta_{FC} - \theta_{WP} \geq 0.05$
Z_r max	Crop range, capped by bed depth sampled from 0.3–1.0 m
Crop	Categorical over the crop table; K_{cb} × uniform(0.9, 1.1); stage lengths × uniform(0.8, 1.2)
	Texture value × log-normal ($\sigma_{\ln} \approx 0.7$). Real K_{sat} varies by an order of magnitude within a

K_{sat}	Texture value × log-normal ($\sigma_{ln} \sim 0.7$). Real K_{sat} varies by an order of magnitude within a texture class.
E_a, f_w	Drip: E_a 0.85–0.95, f_w 0.3–0.4. Sprinkler: E_a 0.65–0.80, f_w 1.0.
Sensor	σ 0.003–0.01, b ±0.03, κ 0–0.001 per °C, drift 0–0.01 per month
Fault onset	Per-hour probabilities, e.g. dropout 1e-3, stuck 1e-4, step 5e-5. These are design choices for stress testing, not measured rates.
Weather	Real historical years from several locations, paired with archived forecasts from the same dates

The sensor and fault ranges are engineering suggestions to make the controller robust, not literature values. Tune them once you have real fault statistics from your hardware.

13 Weather inputs and unit conversions

Open-Meteo variable names, with the conversions the equations above need:

Variable	Open-Meteo field	Native unit	Convert to
Air temperature	`temperature_2m`	°C	as is
Relative humidity	`relative_humidity_2m`	%	as is
Wind speed	`wind_speed_10m`	km/h (default)	÷ 3.6 → m/s,
Shortwave radiation	`shortwave_radiation`	W/m², mean of preceding hour	× 0.0036 → M
Precipitation	`precipitation`	mm, sum of preceding hour	as is
Surface pressure	`surface_pressure`	hPa	÷ 10 → kPa
Reference ET	`et0_fao_evapotranspiration`	mm, preceding hour	as is
Rain probability	`precipitation_probability`	%	forecast only

Timestamp alignment.

Open-Meteo labels hourly sums and means at the *end* of the hour: the 14:00 value covers 13:00–14:00. Align your sensor readings to the same convention, or you'll build in a one-hour shift that looks like a leak or a lag in the model.

14 References

1. Allen, R.G., Pereira, L.S., Raes, D., Smith, M. (1998). *Crop evapotranspiration:*